

**Recording:** The quantity and transfer of assets like containers, pallets, etc, can be recorded as they move between places.

**Sharing:** Information sharing increases due to the transparency of the blockchain. As anyone on the network can see every other transaction happening, the data is reliable.

Also, this reliable data can be used for machine learning models with very high accuracy rates. This will help to predict future needs in a better way and can make the system significantly faster.

## Integrating the supply chain into the blockchain

Blockchain can be applied to each stage of the healthcare supply chain. These include raw material supply, logistics, manufacturing, wholesale distribution and service providers, as well as electronic health records (EHR), validating patient data and even analysing a patient's health records.

At the local level, the staff can maintain inventory as digital records, which will be put on the blockchain. By setting a minimum threshold through smart contracts, orders can be automatically placed for the needed supplies. All types of purchase orders (PO) and delivery orders (DO) can be maintained in a more flexible and accessible manner. Delivery orders can be pushed to logistics and the process will start. Also, we can tag shipments with bar codes and store them on the blockchain, against the delivery order form. This bar code can then be used to easily track the source of the package, leading to increased efficiency.

In hospitals, patients' data is already stored electronically as EHR (electronic health records). But trust and transparency between organisations has always been an issue. By providing an immutable ledger with limited access to data, the blockchain can solve this problem. Data once recorded cannot be deleted and will always be there to look up. This helps doctors to have a deep insight into a patient's health, which results in better diagnoses.

Prescription medication can be put on the blockchain against a customer's account which will be visible to medical stores. The medical store, which is

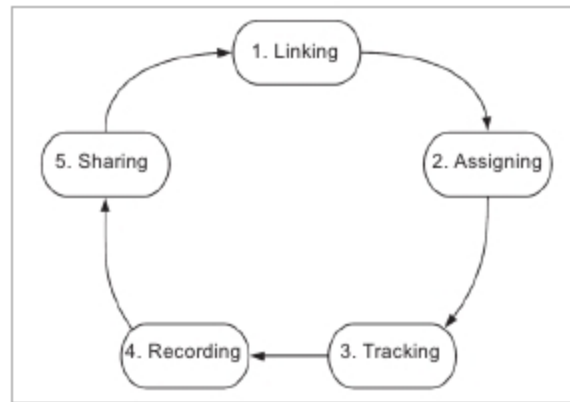


Figure 6: Sequential flow of the supply chain and the advantages integrated with blockchain

already assured of the legitimacy of the prescription, can inform the patient about the availability of medicines required. In this way, the blockchain solves problems such as forged prescriptions and double buying, and keeps a record of the final consumer of the medicine.

For medical giants, raw materials supply can be recorded and put onto a manufacturer's blockchain network. The quantity of produced supplies and disposable goods can also be recorded digitally. Pending orders can be settled using time based escrow contracts, eliminating the need for a third party to approve the orders. Timestamps enable the blockchain to perform any transaction in a time sequential manner.

Also, the monetary value of reliable health records is very high for insurance companies. Access to the trusted data of the patient can help insurance companies decide on a better premium, which will increase the claim settling rates of the health insurance industry. The automation achieved can provide faster claim settlements, increasing the confidence of the customer towards insurance companies.

With the digitisation and automation achieved, the blockchain architecture will also fulfil the following objectives.

1. Enhanced transparency, as end to end details are stored and accessible.
2. Scalability: Any number of participants from any touchpoint can be added.

3. Better security: The data, being digital and immutable with no centralised access, cannot be tampered with.

4. Increased innovation: As data stored from every stage has been linked in a decentralised form, it can be used to predict supplies, thus removing loopholes and increasing innovation.

All these enhanced features of the blockchain maximise the efficiency and also strengthen the relationship across the supply chain.

## Proof of identity and security

Every user on the blockchain has a private and a public key. Any time a user wants to do some transaction, he or she creates a digital signature using a private key along with the data to be sent. So this newly created digital signature will be sent along with the transaction to the receiver. The receiver will have the sender's public key and using that, the user can be verified—if the digital signature was created using the private key behind the sender's public key.

If anyone tries to tamper with the transaction in between, the sender's public key will not be able to verify the new digital signature created using the sender's private key as data has changed. This way the sender is provided security even if the data stored in form of encrypted hash is public.

To add people to the network, a POI (proof of identification) is needed. India has already implemented and adopted Aadhaar, an identification number. Services such as e-KYC can be used to authorise a new user onto the network. Aadhaar also directly connects a person to his/her bank account, which leads to less overhead for insurance companies.

The principle behind the blockchain is to make transactions public, but this is a big concern for health records. In our next article, we will explain how to encrypt data, when using blockchain for governance and also discuss the level controls of the data. **END** 🐧

By: Vaibhav Aggarwal and Prof. B. Thangaraju

The authors are associated with the open source technology lab in the International Institute of Information Technology, Bengaluru. Aggarwal can be reached at [vaibhav.aggarwal@fidesgc.com](mailto:vaibhav.aggarwal@fidesgc.com).